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Model Analysis

Four comparisons had to be made throughout the whole lab. Comparison between the six models: logistic regression, decision tree, random forest, gradient boosting, k-neighbor classifier, and support vector machine (svm). Two comparisons with ensemble models, the voting classifier ensemble and the average ensemble models. Both models were used to compare accuracy, precise, the recall, their f1 score, and roc-auc in validation and test data. Last comparison was between the model of my choice, the voting classifier ensemble model, and the Bayesian model.

However, I will be looking into the comparison of the voting classifier and the average ensemble models. The way to evaluate the models isn't too different but each contains their key code for each model to function properly. Despite this, the results that each gave in the validation set of data were equal. As for the test dataset results, both differed from the validation dataset but remained equal to each other. Even though the strategies to carry out the outcome differed, both models displayed equal metrics: equal accuracy, equal precision, equal recall, equal f1 score, and an equal roc-auc score. By being equal, it lets us know that the training data might've been similar, which was, but it also shows that the models could be highly aligned or that there wasn't

enough diversity present. Given all of this, the conclusion we get is that the models, in this case, are not better or worse than one another.

The approach was the same: Define the model, train the ensemble, define predictions and probabilities on validation dataset, evaluate on validation dataset, define predictions and probabilities on test dataset, evaluate on the test dataset.

After having both models go through that process, a comparison had to be made. First the 'ensemble_comparison' variable was created with the function 'pd.DataFrame' and within its variables, the model, accuracy, precision, recall, f1 score, and roc-auc; this was repeated once more in order to adapt it to the test dataset with now the variable being 'test_ensemble_comparison' followed by 'pd.DataFrame' as well. The variables within the 'pd.DataFrame' function were changed to the adequate ones that were used for the test dataset for accuracy, precision, etc.